

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 02 · LECTURE NOTES

Games and Equilibria

The solution concepts, and what Nash's theorem does not promise.

MODULE OUTCOMES

- Define a game in normal form.
- Define dominance, Nash, and the relations between them.
- State Nash's theorem precisely and its three caveats.
- Explain mixed strategies and what they model.
- Analyse small games correctly.

1 The objects

PLAYERS, STRATEGIES, PAYOFFS — AND EVERY MODELLING DECISION IS IN THE PAYOFFS

A normal-form game is a set of players, a strategy set for each, and a payoff function mapping strategy profiles to a real number per player — three objects, and the first two are usually uncontroversial.

And the payoffs encode everything you believe about what the agents want — which is the modelling step, and is where essentially all disagreements about a game-theoretic analysis actually live, though they are frequently conducted as disagreements about the solution concept.

So 'the prisoner's dilemma shows cooperation is irrational' is a claim about a particular payoff matrix, not about people — and the matrix was chosen by somebody, with the assumption that the players care about nothing except their own sentence built into it.

Which is this course's first caution: argue about the payoffs before arguing about the equilibrium, because the equilibrium follows mechanically from the payoffs and the payoffs do not follow from anything — they are an assumption.

2 Solution concepts

Concept	Definition	What it assumes about the agents
Dominant strategy equilibrium	Each player's strategy is best regardless of what others do.	Nothing at all about beliefs. Rare, and ideal — see the note.

Concept	Definition	What it assumes about the agents
Iterated dominance	Remove strictly dominated strategies, repeat.	Common knowledge of rationality — everyone is rational, everyone knows that, and so on.
Pure Nash equilibrium	No single player gains by deviating alone.	Correct beliefs about the others. May not exist.
Mixed Nash equilibrium	The same, allowing randomised strategies.	Correct beliefs. Always exists in a finite game (§3).
Correlated equilibrium	Obey a recommendation from a shared signal.	A coordinating device, and that obeying is best given the signal — and it is what no-regret dynamics converge to (Module 12).

A dominant strategy equilibrium needs no assumption whatsoever about what the other agents do — which is exactly why mechanism design aims for it (Module 06 §2) rather than for Nash. The assumption column is why dominant strategies are the design target: a mechanism whose guarantee requires every participant to correctly model every other participant is guaranteeing very little in practice.

AND NASH EQUILIBRIUM ASSUMES AGENTS HAVE CORRECT BELIEFS ABOUT EACH OTHER, WHICH IS A LOT

A Nash equilibrium is a strategy profile in which no player gains by deviating alone — which presupposes that each agent correctly anticipates the strategies the others are playing, since otherwise 'gains by deviating' is evaluated against the wrong thing.

So it is a consistency condition rather than a prediction of behaviour — it says what is stable once reached, not how anybody would get there — which is Module 12's question and is a genuinely different one.

And games may have many equilibria, with no principle within the concept itself that selects among them — which is a serious limitation for anybody who wanted a prediction, and has generated a large literature on refinements that mostly did not settle it.

Which is why this course prefers dominant strategies wherever it can get them — they require no belief about anybody — and a mechanism that relies on correct mutual beliefs is relying on a great deal that the designer cannot verify.

3 Nash's theorem

Every finite game has at least one mixed Nash equilibrium.

- **Finitely many players, finitely many strategies each** — proved by a **fixed-point argument** (Brouwer or Kakutani), **which is why Module 03's complexity result has the shape it does**: the problem is complete for a class defined by exactly that kind of existence proof.
- **It does not say the equilibrium is unique, or that it is good for anybody** (Module 04's price of anarchy), **or that agents will find it** (Module 12), **or that it can be computed efficiently** (Module 03) — **four separate questions, and three of them have discouraging answers.**
- **And 'mixed' means randomising, which has to be interpreted: as deliberate randomisation by a player, as a frequency within a population of players, or as the other players' uncertainty about what this one will do** — **three readings, and the right one depends entirely on the application**, which is worth settling before quoting a mixed equilibrium at anybody.
- **Existence is not uniqueness, quality, reachability, or computability** — **and the last three are Modules 03, 04, and 12 respectively. The four caveats structure the next three modules**, which is a convenient way to remember what each is for.

4 Reading a game

- **Look for dominant strategies first** — **if one exists, the analysis is finished**, the prediction is robust, and no assumption about beliefs is needed.
- **Then iterated removal of strictly dominated strategies** — **which often shrinks the game dramatically and sometimes solves it outright**, and is cheap to do.
- **Then pure Nash by inspection: check every cell against each player's possible deviations** — mechanical, tedious, and it catches the errors that cleverness introduces.
- **Then mixed equilibria, by making the opponent indifferent** — **which is the standard technique and is worth deriving once from the definition** rather than memorising: a player only randomises if the strategies being mixed give equal expected payoff, so the *other* player's mixture must be the one that equalises them.
- **And then ask whether the payoffs were right (§1)** — **which is the step that is never taken and matters most. In a mixed equilibrium each player randomises so as to make the *other* indifferent** — **which is counterintuitive and is the key to computing them.**

AND A SOLUTION CONCEPT IS A MODELLING CHOICE, WHICH SHOULD BE NAMED

'The equilibrium of this game is X' means 'under this solution concept, with these payoffs' — and both qualifiers do real work, which is why both belong in the sentence.

Which matters because the concepts genuinely disagree: a game can have a unique correlated equilibrium and several Nash equilibria, or a dominant strategy that iterated dominance would also find but that a casual Nash analysis obscures — and the predictions differ.

So name the concept, as you would name a statistical model or a distributional assumption — which is CSCE 628 Module 13's methods discipline arriving in a purely theoretical setting, and is the same habit.

And prefer the concept with the weakest assumptions that still answers your question — which is almost always dominant strategies when you are the designer rather than the analyst (Module 06 §2), because you get to choose the game.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Shoham & Leyton-Brown, chapters 3 and 4 (free PDF)	§§1 to 3 — the solution concepts, carefully distinguished, which is what this module is about.
Nash — Equilibrium points in n-person games (free)	§3 — one page, and the proof is the fixed-point argument in its original form.
Osborne & Rubinstein — A Course in Game Theory (free PDF from the authors)	§§2 and 4 — rigorous, free, and unusually careful about what each concept assumes.
The Gambit game solver (free)	§4 — compute equilibria of small games, and watch it become infeasible, which is Module 03's point made experimentally.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Write three games in normal form, with the payoffs justified.
- 2 Change one payoff in the prisoner's dilemma so that cooperation is dominant, and say what you assumed.
- 3 Find the dominant strategies in five games, or show there are none.
- 4 Solve a game by iterated dominance.

- 5 **Find all pure Nash equilibria** of a 3 by 3 game by inspection.
- 6 **Find a game with no pure equilibrium** and compute its mixed one.
- 7 **Derive the indifference condition** from the definition.
- 8 **Find a game with multiple equilibria** and say which one you would predict, and why.
- 9 **Run a solver** on games of increasing size and record where it stops.
- 10 **Take a published game-theoretic claim** and identify its payoff assumptions.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the three components of a game.
- 2 Where does the modelling happen, and why does that matter?
- 3 Name five solution concepts and what each assumes.
- 4 Why are dominant strategies the design target?
- 5 What does a Nash equilibrium presuppose?
- 6 Why is it a consistency condition rather than a prediction?
- 7 State Nash's theorem precisely.
- 8 Give its four caveats and which modules address them.
- 9 Give three interpretations of a mixed strategy.
- 10 Give the analysis order for a small game.